Datos del problema

VG=24 V
L=10 mH
C=10 mF
fs=1 kHz
D=0.45

1. Obtención analítica de Vo

$VL = VG - Vo$ $\qquad$ $ton = DT$

$VL = \frac{VG}{2} - Vo$ $\qquad$ $toff = (1 - D)T$

$(VG - Vo)DT$

$\left(\frac{VG}{2} - Vo\right)(1 - D)T$

$$\frac{(VG - Vo)DT + \left(\frac{VG}{2} - Vo\right)(1 - D)T = 0}{T}$$

$$(VG - Vo)D + \left(\frac{Vo}{2} - Vo\right)(1 - D) = 0$$

$$DVG - DVo + \frac{VG}{2} - Vo - \frac{DVG}{2} + VoD = 0$$

$\frac{VG}{2}(1 - D) - Vo$ =0

$$\frac{VG}{2}(1 - D) = Vo$$

$$Vo = \frac{Vg}{2}(1 + D)$$

$$Vo = \frac{24}{2}(1 + 0.45)$$

$$Vo = 12(1.45)$$

$$Vo = 17.4V$$

3. Corriente promedio de salida Io

$$Io = \frac{Vo}{Ro} = \frac{\Delta IL}{2}$$

$$\Delta IL = \frac{(VG - Vo)D}{Lfs}$$

$$\frac{(24 - 17.4)0.45}{10x10^{-3} - 1x10^{3}}$$

$$\frac{6.6(0.45)}{10} = \frac{2.92}{10}$$

$$\Delta IL = 0.292\text{A}$$

4. Resistencia crítica

$$Io = \frac{\Delta IL}{2}$$

$$\frac{Vo}{Rc} = \frac{\Delta IL}{2}$$

$$2Vo = Rc\Delta IL \therefore \frac{2Vo}{\Delta IL} = Rc$$

$$Rc = \frac{2(17.4)}{0.247}$$

$$Rc = 117.17\Omega$$

5. Selección de la Resistencia

$ILmin > 0\ CCM \qquad ILmin = 0\ DCM$

$$Io = \frac{\Delta IL}{2} = \frac{0.297}{2} = 0.1485A\ > 0\ CCM$$

$$Rc > R \qquad R = 100\Omega$$

6. Rizado de corriente

$$\Delta iL = \frac{17.4}{100} = 0.174A$$

7. Rizado de tensión

$$\Delta Vo = \frac{\Delta IL}{8fsC} = \frac{0.297}{8(1x10^{3})(10x10^{-3})} = \frac{0.292}{80} = 3.712mV$$

8. Porcentaje de rizado

$$\%\Delta Vc = \frac{\Delta Vo}{Vo} x100$$

$$\frac{3.7x10^{-3}}{17.4} x100$$

$$\%\Delta Vc = 0.0213\%$$

$$T = \frac{1}{fs} = \frac{1}{1000} = 1X10^{-3}$$

$$tan = DT = 0.45(1X10^{-3}) = 4.5x10^{-4}$$

$$= 450u$$

14. Error porcentual

Variable	Valor teórico	Valor simulado	Error (%)
Vo	17.4v	11.4	-34.09
ΔiL	0.292	0	0.292
ΔVo	3.712	357	35.6
	0.0213	357	-16.7
Rizado (%)			